Practice workbook · Describe virus structure and reproduction

The protein coat and nucleic acid of viruses and how they reproduce inside host cells.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VIRUS-STRUCTURE-REPRODUCTION

1. Question 9 · part b · 6 marks

June 2022 Paper 2

Worked-example retry · Attempt part b only.

9 (a) Hepatitis is a disease of the liver.

This disease can be caused by drinking too much alcohol.

Explain the damaging effects of alcohol on the liver.

(3)

(b) Hepatitis B is one type of virus that can cause hepatitis.

Describe the reproduction of a hepatitis B virus.

(6)

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1. Question 9 · part b · 6 marks

June 2022 Paper 2

Worked-example retry · Attempt part b only.

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(c) Vaccinations can be used to prevent the spread of hepatitis.

Explain how vaccines can help people become immune to hepatitis.

(3)

(Total for Question 9 = 12 marks)

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1. Question 9 · part b · 6 marks · Working and answer

Working and answer

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2. Question 2 · part a.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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2. Question 2 · part a.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

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2. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 8 · part a.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.i only.

8 (a) (i) Describe the structure of a virus.

(2)

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

(5)

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3. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

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